

ESAT PRACTICE TEST SERIES

MATHS 2 + CHEMISTRY

LEVEL 1 • STANDARD

Maths 2 + Chemistry

Mathematics 1 + Mathematics 2 + Chemistry

Representative ESAT-style breadth with routine and multi-step applications.

MODULE

Mathematics 1

MODULE

Mathematics 2

MODULE

Chemistry

FULL SOLUTION BOOK

ThrivingScholars

81 preserved questions • three verified module keys

Maths 2 + Chemistry • Level 1

Standard

This book compiles three existing 27-question modules into one 81-question pathway. No new questions have been generated and no source solution has been rewritten.

Composition: Mathematics 1 + Mathematics 2 + Chemistry. Each module retains its original question order, answer and worked solution.

Mathematics 1

Engineering Mock Test 3 · preserved Mathematics 1 module

Q1 C	Q2 B	Q3 E	Q4 C	Q5 D
Q6 D	Q7 E	Q8 A	Q9 A	Q10 D
Q11 D	Q12 C	Q13 B	Q14 A	Q15 B
Q16 E	Q17 E	Q18 E	Q19 D	Q20 E
Q21 D	Q22 D	Q23 C	Q24 D	Q25 D
Q26 A	Q27 C			

Mathematics 2

Engineering Mock Test 3 · preserved Mathematics 2 module

Q1 C	Q2 A	Q3 C	Q4 C	Q5 C
Q6 B	Q7 D	Q8 E	Q9 D	Q10 F
Q11 D	Q12 B	Q13 A	Q14 D	Q15 E
Q16 G	Q17 D	Q18 G	Q19 C	Q20 C
Q21 C	Q22 G	Q23 E	Q24 D	Q25 A
Q26 B	Q27 A			

Chemistry

Preserved Chemistry Standard source set

Q1 C	Q2 D	Q3 B	Q4 C	Q5 E
Q6 B	Q7 E	Q8 C	Q9 D	Q10 D
Q11 E	Q12 F	Q13 B	Q14 E	Q15 C
Q16 B	Q17 D	Q18 C	Q19 A	Q20 C
Q21 B	Q22 C	Q23 D	Q24 C	Q25 A
Q26 E	Q27 C			

Mathematics 1

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 1 module. Question wording, option order, answer and solution method are unchanged.

M1-01

Mock Test 3 - Mathematics 1

Answer **C****QUESTION 1**

There are 8 consecutive integers. If the sum of the first three is 63, what is the largest one?

A 17**B** 25**C** 27**D** 30**E** 31**WORKED SOLUTION**

Let the first integer be n . Then $n + (n + 1) + (n + 2) = 63$, so $3n + 3 = 63$, giving $n = 20$. The largest of the eight integers is 27.

QUESTION 2

Joshua is 10 years old and his father is 42 years old. After how many years, his father's age is three times Joshua's age?

A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

After t years, $42 + t = 3(10 + t)$. Hence $42 + t = 30 + 3t$, so $t = 6$.

QUESTION 3

N is a 7-digit number consists of 2 or 3. There are more 2s than 3s in N . If N is divisible by both 3 and 4, what is the remainder of N when it is divided by 7?

A 1

B 2

C 3

D 4

E 5

WORKED SOLUTION

Let k be the number of 3s. The digit sum is $14 + k$, so divisibility by 3 gives $k = 1$. Divisibility by 4 forces the last two digits to be 32. Thus $N = 2222232$, and $N \equiv 5 \pmod{7}$.

M1-04

Mock Test 3 - Mathematics 1

Answer C

QUESTION 4

The preliminary round of a violin competition is divided into four rounds. According to the rule, the average score of a percipient of these four rounds must be not less than 96 points in order to enter the finals. Shaelyn's scores in the first three rounds are 95, 97, and 94. What is the minimum score that she needs to get on the fourth round in order to enter the finals? (Hope Cup Math Competition, China)

A 96

B 97

C 98

D 99

E 100

WORKED SOLUTION

The total score needed is $4 \times 96 = 384$. The first three scores total $95 + 97 + 94 = 286$, so the fourth score must be at least $384 - 286 = 98$.

M1-05

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 5

A haunted house has 8 windows. In how many ways can George the Ghost enter the house by one window and leave by a different window?

A 28

B 32

C 40

D 56

E 63

WORKED SOLUTION

There are 8 choices for the entry window and then 7 different choices for the exit window, giving $8 \times 7 = 56$.

M1-06

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 6

Jackie bought a bottle (30 oz) of 90% alcohol from pharmacy to make homemade disinfectant spray. How many ounces of water should he add in order to dilute it into 75% alcohol? (Hint: The amount of alcohol remains the same.)

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

The amount of alcohol is $30 \times 0.90 = 27$ oz. If the final concentration is 75%, then $0.75T = 27$, so $T = 36$. Water added: $36 - 30 = 6$ oz.

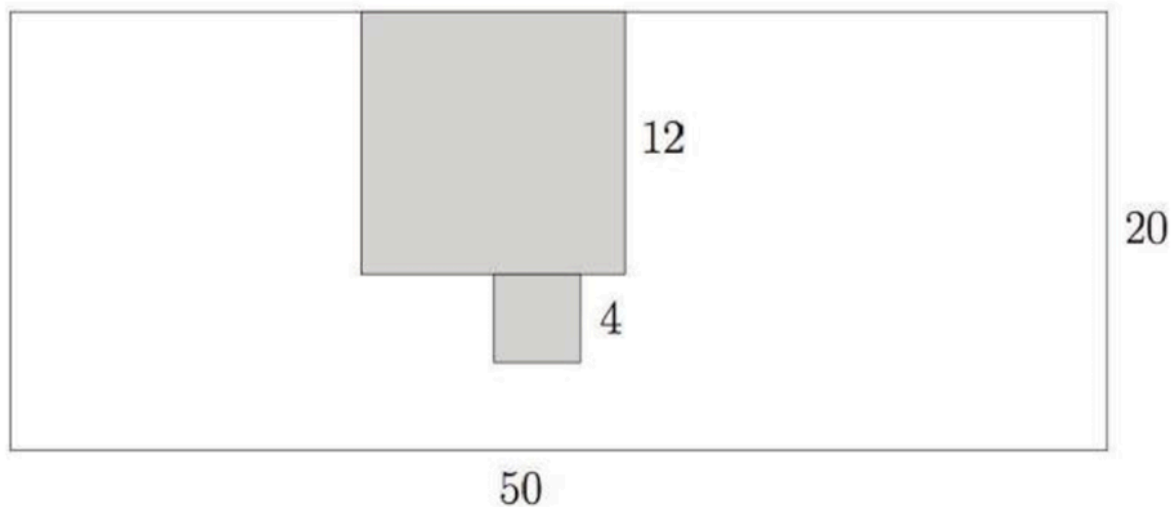
M1-07

Mock Test 3 - Mathematics 1

Answer E

QUESTION 7

Cut a 12×12 square and a 4×4 square from a 50×20 rectangle paper. What is the perimeter of the remaining part?



A 128

B 140

C 152

D 160

E 172

WORKED SOLUTION

The original rectangle perimeter is $2(50 + 20) = 140$. The 12×12 notch increases the perimeter by 24, and the attached 4×4 cut-out adds another 8. Total perimeter = $140 + 32 = 172$.

M1-08

Mock Test 3 - Mathematics 1

Answer A

QUESTION 8

Jasmine keeps adding numbers one by one from the following list

1, 3, 5, 7, ...

as $1 + 3 = 4$, $1 + 3 + 5 = 9$, $1 + 3 + 5 + 7 = 16$, etc. At one point, she got 379 and realized that one number is missed in the addition. What is the sum of digits of the missing number?

A 3

B 6

C 8

D 9

E 11

WORKED SOLUTION

The sum of the first n odd numbers is n^2 . Since 379 is just below $20^2 = 400$, the missing number is $400 - 379 = 21$. The digit sum is $2 + 1 = 3$.

M1-09

Mock Test 3 - Mathematics 1

Answer A

QUESTION 9

$\overline{ab}$ and $\overline{cd}$ are two 2-digit numbers. If

$$\overline{ab} - \overline{cd} = 24$$

and

$$\overline{1ab} - \overline{cd1} = 15,$$

what is $a + d$?

A 5

B 8

C 10

D 11

E 12

WORKED SOLUTION

Using place value, $\overline{ab} = 10a + b$ and $\overline{cd} = 10c + d$. Solving the two given equations gives $a + d = 5$.

M1-10

Mock Test 3 - Mathematics 1

Answer D

QUESTION 10

The speed of a train is 15 meters per second. If it takes the train 10 seconds to pass a lamp post, how long does it take the train to completely pass a 300-meter long tunnel, in seconds? (Hint: first figure out the length of the train based on the information related to the lamppost)

A 10

B 20

C 25

D 30

E 35

WORKED SOLUTION

The train length is $15 \times 10 = 150$ m. To pass the tunnel completely, it travels $300 + 150 = 450$ m. Time = $450 / 15 = 30$ s.

M1-11

Mock Test 3 - Mathematics 1

Answer D

QUESTION 11

How many pairs of positive integers (x, y) satisfy that $\frac{x}{3} = \frac{4}{y}$?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

From $x/3 = 4/y$, we get $xy = 12$. The positive factor pairs are (1, 12), (2, 6), (3, 4), (4, 3), (6, 2), (12, 1), so there are 6 pairs.

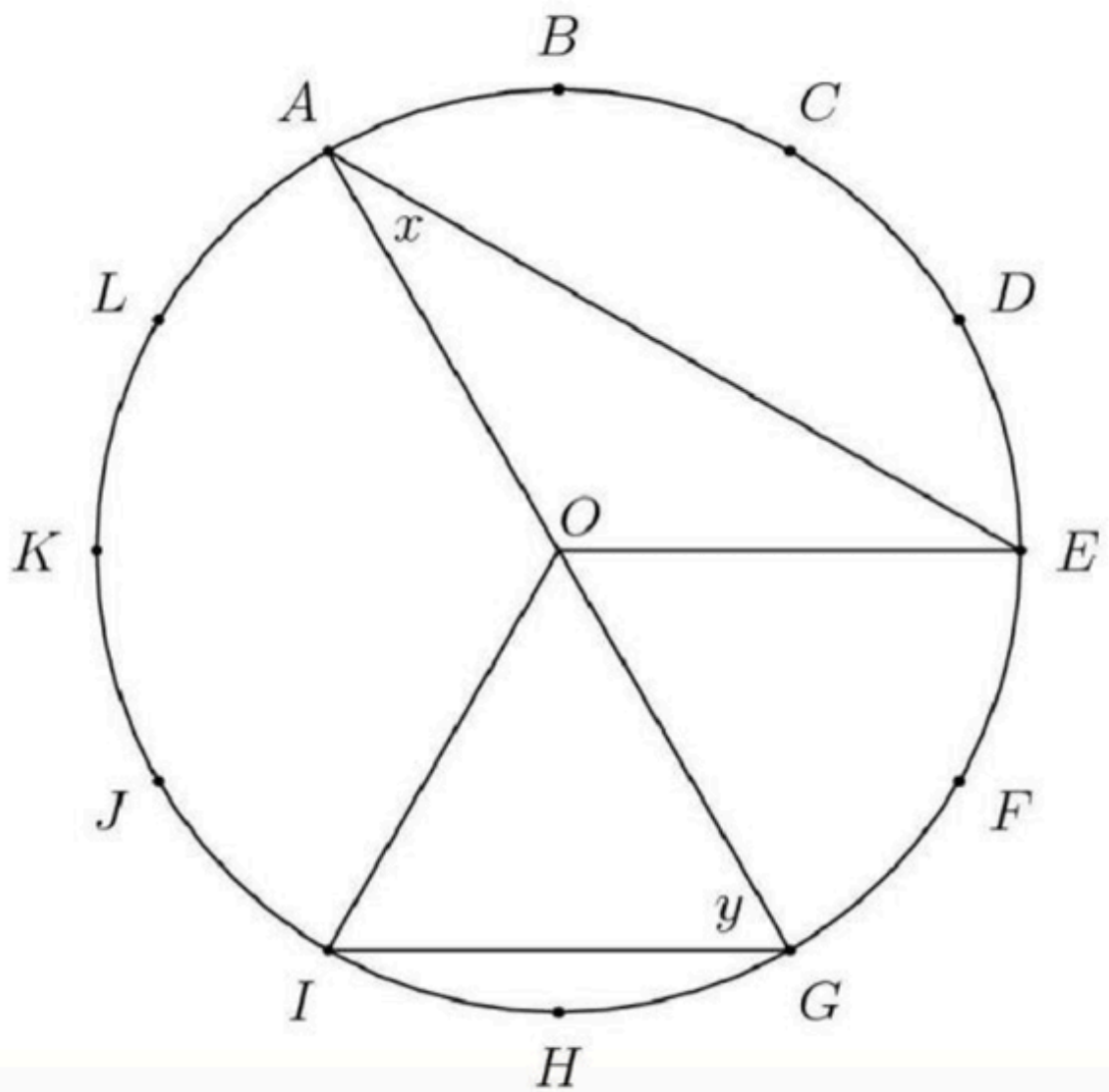
M1-12

Mock Test 3 - Mathematics 1

Answer **C**

QUESTION 12

The circumference of the circle with center O is divided into 12 equal arcs, marked the letters A through L as seen below. What is the number of degrees in the sum of the angles x and y ?



A 75

B 80

C 90

D 120

E 150

WORKED SOLUTION

The circle is divided into 12 equal arcs, so each arc is 30° . Using central and inscribed angle relationships from the diagram gives $x + y = 90^\circ$.

M1-13

Mock Test 3 - Mathematics 1

Answer **B**

QUESTION 13

If for whole numbers x , y and z ,

$$360 = 2^x \times 3^y \times 5^z,$$

what is $x + y + z$? Here $a^n = a \times a \times \dots \times a$.

For example, $100 = 2 \times 2 \times 5 \times 5 = 2^2 \times 5^2$.

A 5

B 6

C 7

D 8

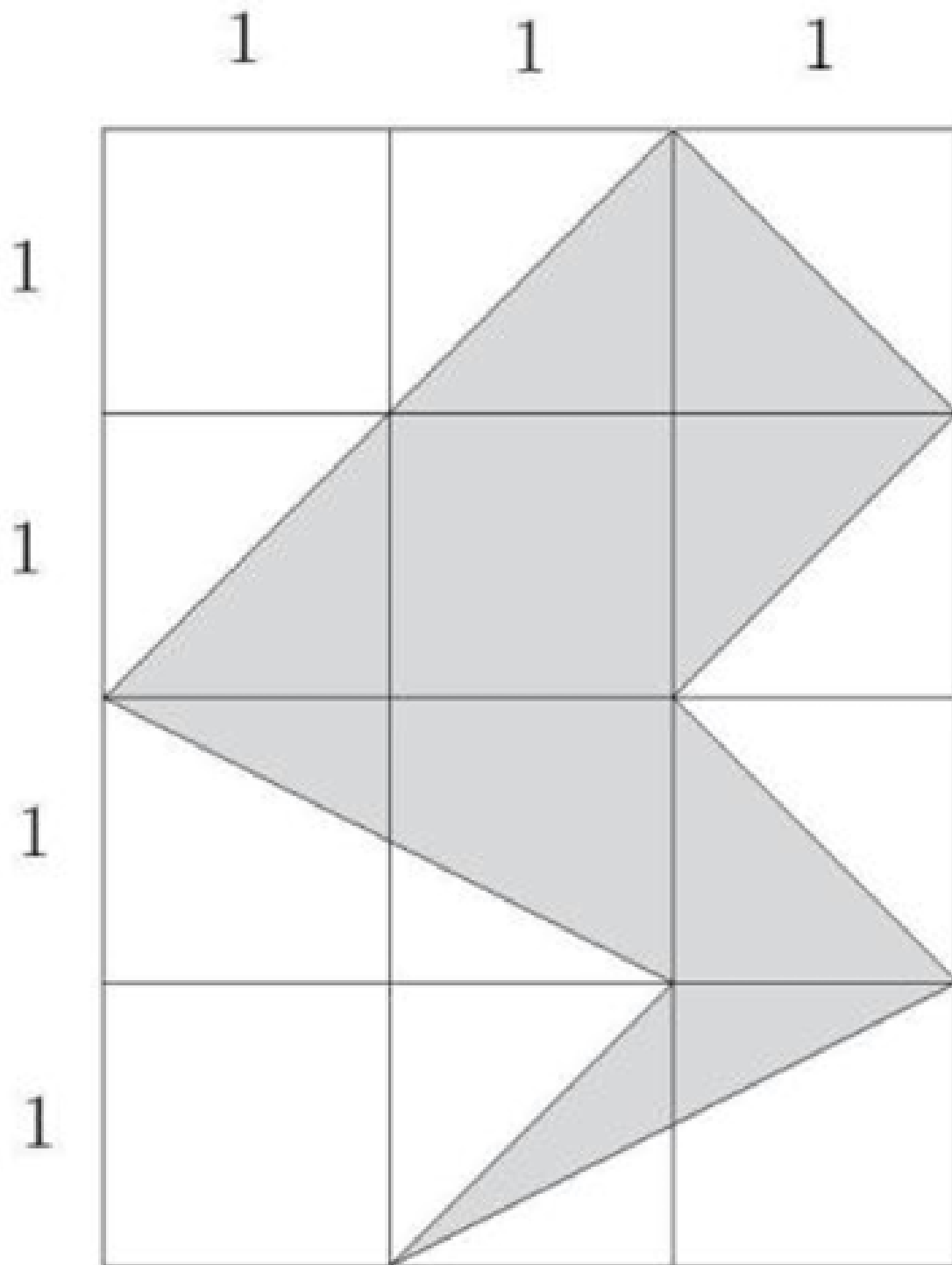
E 9

WORKED SOLUTION

Prime factorise: $360 = 36 \times 10 = 2^2 \times 3^2 \times 2 \times 5 = 2^3 \times 3^2 \times 5^1$. Hence $x + y + z = 3 + 2 + 1 = 6$.

QUESTION 14

What is the area of the shaded region in the following 3×4 grid?



A 5

B 6

C 7

D 8

E 9

WORKED SOLUTION

Counting the shaded polygon by subtracting the surrounding right triangles from the 3×4 rectangle gives shaded area 5.

M1-15

Mock Test 3 - Mathematics 1

Answer **B**

QUESTION 15

Randomly choose 3 different numbers from the set $\{1, 3, 5, 7, 9\}$. What is the probability that the three selected numbers could be side lengths of a triangle?

A $\frac{1}{5}$

B $\frac{3}{10}$

C $\frac{1}{3}$

D $\frac{2}{5}$

E $\frac{1}{2}$

WORKED SOLUTION

There are $\binom{5}{3} = 10$ triples. The valid triangle triples are $(3, 5, 7), (3, 7, 9), (5, 7, 9)$, so the probability is $3/10$.

M1-16

Mock Test 3 - Mathematics 1

Answer E

QUESTION 16

Marley practices exactly one sport each day of the week. She runs three days a week but never on two consecutive days. On Monday she plays basketball and two days later golf. She swims and plays tennis, but she never plays tennis the day after running or swimming. Which day of the week does Marley swim?

A Sunday**B** Tuesday**C** Thursday**D** Friday**E** Saturday**WORKED SOLUTION**

Monday is basketball and Wednesday is golf. Applying the non-consecutive running rule and the condition that tennis is not immediately after running or swimming leaves Saturday as the swimming day.

QUESTION 17

Three toy cars run around a round track. It takes them 8, 10 and 12 seconds to finish one full circle respectively. If three cars start together from the same point and run toward the same direction, after how many seconds will they get back to the starting point together for the first time?

A 20

B 40

C 60

D 80

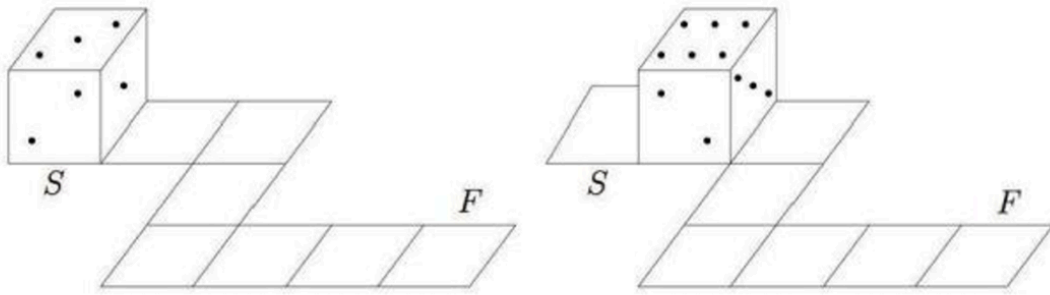
E 120

WORKED SOLUTION

The cars return together after the least common multiple of 8, 10, 12, which is 120 seconds.

QUESTION 18

Opposite faces of a die always add to 7. A die rolls on a circuit as represented below. At the starting point (S) the top face is 3. Which will be the top face at the final point (F)?



A 2

B 3

C 4

D 5

E 6

F 7

G 8

H 9

WORKED SOLUTION

Track the die through the path, using opposite faces summing to 7 after each roll.
The final top face is 6.

QUESTION 19

How many numbers x satisfy that $(x^2 - 2)^2 = 4$?

A 0

B 1

C 2

D 3

E 4

F 5

G 6

H 7

WORKED SOLUTION

$(x^2 - 2)^2 = 4$ gives $x^2 - 2 = \pm 2$. Thus $x^2 = 4$ or $x^2 = 0$, giving $x = -2, 0, 2$. There are 3 real numbers.

QUESTION 20

For how many two-digit numbers, if you switch the tens and ones digits (e.g., $18 \rightarrow 81$), the new 2-digit number is 18 more than the original one?

A 3

B 4

C 5

D 6

E 7

WORKED SOLUTION

Let the number be $10a + b$. Reversing digits gives $10b + a$. The condition gives $10b + a = 10a + b + 18$, so $b - a = 2$. There are 7 possible tens digits.

M1-21

Mock Test 3 - Mathematics 1

Answer D

QUESTION 21

[AMC 8 Problem] Malcolm wants to visit Isabella after school today and knows the street where she lives but doesn't know her house number. She tells him, "My house number has two digits, and exactly three of the following four statements about it are true."

- It is prime.
- It is even.
- It is divisible by 7.
- One of its digits is 9.

This information allows Malcolm to determine Isabella's house number. What is its units digit?

A 4

B 6

C 7

D 8

E 9

WORKED SOLUTION

The number must have exactly three of the four properties. The number 98 is even, divisible by 7, has a digit 9, and is not prime. Its units digit is 8.

M1-22

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 22

In the addition problem shown, m , n , p , and q represent positive digits. When the problem is completed correctly, the value of $m + n + p + q$ is (Canadian Math Competition)

t is prime.

t is even.

t is divisible by 7.

One of its digits is 9.

information allows N

A 16

B 18

C 22

D 24

E 30

WORKED SOLUTION

From the units column, $3 + 2 + q$ ends in 2, so $q = 7$ with carry 1. From the tens column, $6 + p + 8 + 1$ ends in 4, so $p = 9$. From the hundreds column,

$n + 7 + 5 + 2$ ends in 0, so $n = 6$ with carry $m = 2$. Therefore
 $m + n + p + q = 2 + 6 + 9 + 7 = 24$.

M1-23

Mock Test 3 - Mathematics 1

Answer C

QUESTION 23

If two numbers x and y satisfy that $(x + 1)^2 + (y - 5)^2 = 0$, what is $x + y$?

A -1

B 0

C 4

D 6

E 7

WORKED SOLUTION

A sum of squares is zero only if both squares are zero. Thus $x + 1 = 0$ and $y - 5 = 0$, so $x = -1$, $y = 5$, and $x + y = 4$.

M1-24

Mock Test 3 - Mathematics 1

Answer D

QUESTION 24

Abel tosses two coins and Ben tosses one coin. What is the probability that they get the equal number of heads?

Recall that

$$\text{Probability} = \frac{\text{Number of valid cases}}{\text{Number all possible cases}}.$$

A $\frac{1}{8}$

B $\frac{1}{4}$

C $\frac{1}{3}$

D $\frac{3}{8}$

E $\frac{2}{5}$

WORKED SOLUTION

There are 8 equally likely outcomes. Equal heads occurs when Abel gets 0 heads and Ben gets 0 heads, or when Abel gets 1 head and Ben gets 1 head:

$1 + 2 = 3$ cases. Probability = $3/8$.

M1-25

Mock Test 3 - Mathematics 1

Answer **D**

QUESTION 25

x , y , z and w are four numbers. If the average of the following three numbers

$x - y$, $y - z$, $z - w$ is 10 and $w = 5$. What is x ?

A 15

B 20

C 25

D 35

E there is no enough information

WORKED SOLUTION

The average of $x - y$, $y - z$, $z - w$ is 10, so their sum is 30. The sum telescopes: $(x - y) + (y - z) + (z - w) = x - w = 30$. Since $w = 5$, $x = 35$.

M1-26

Mock Test 3 - Mathematics 1

Answer A

QUESTION 26

What is the tens digit of 7^{2020} ?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

The last two digits of powers of 7 cycle: 07, 49, 43, 01. Since 2020 is divisible by 4, 7^{2020} ends in 01, so the tens digit is 0.

QUESTION 27

Two trains 300 miles apart are traveling toward each other along the same track. One train goes 70 miles per hour; the other runs at 80 miles per hour. If both of them leave their stations at 7:00am, when will they meet?

A 8 : 00 am

B 8 : 30 am

C 9 : 00 am

D 9 : 30 am

E 10 : 00 am

WORKED SOLUTION

The trains approach each other at $70 + 80 = 150$ miles per hour. Time
 $= 300 / 150 = 2$ hours, so they meet at 9:00 am.

Mathematics 2

27 QUESTIONS • COMPLETE SOLUTIONS

Engineering Mock Test 3 · preserved Mathematics 2 module. Question wording, option order, answer and solution method are unchanged.

M2-01

Mock Test 3 - Mathematics 2

Answer C

QUESTION 1

Rearrange $y^4 - 4y^3 + 6y^2 - 4y + 2 = x^5 + 7$ to make y the subject, taking the principal fourth-root branch.

A $y = 1 + (x^5 + 7)^{1/4}$

B $y = -1 + (x^5 + 7)^{1/4}$

C $y = 1 + (x^5 + 6)^{1/4}$

D $y = -1 + (x^5 + 6)^{1/4}$

WORKED SOLUTION

Complete the binomial pattern:

$$y^4 - 4y^3 + 6y^2 - 4y + 1 = (y - 1)^4.$$

So $(y - 1)^4 + 1 = x^5 + 7$, hence $(y - 1)^4 = x^5 + 6$. Taking the principal fourth root gives $y = 1 + (x^5 + 6)^{1/4}$.

QUESTION 2

The aspect ratio of my television screen is 4:3 and the diagonal is 50 inches. What is the area of my television screen?

A 1,200 inches²

B 1,000 inches²

C 120 inches²

D 100 inches²

E More information needed.

WORKED SOLUTION

Question 2: A

Let the width of the television be $4x$ and the height be $3x$.

By Pythagoras,

$$(4x)^2 + (3x)^2 = 50^2$$

Simplify:

$$25x^2 = 2500 \Rightarrow x = 10$$

Therefore, the screen is 30 inches by 40 inches, so the area is 1200 inches².

QUESTION 3

Rearrange the equation $\sqrt{1 + 3x^{-2}} = y^5 + 1$ to make x the subject, taking the positive square-root branch.

A $x = \frac{y^{10} + 2y^5}{3}$

B $x = \frac{3}{y^{10} + 2y^5}$

C $x = \sqrt{\frac{3}{y^{10} + 2y^5}}$

D $x = \sqrt{\frac{y^{10} + 2y^5}{3}}$

E $x = \sqrt{\frac{y^{10} + 2y^5 + 2}{3}}$

F $x = \sqrt{\frac{3}{y^{10} + 2y^5 + 2}}$

WORKED SOLUTION

Square both sides:

$$1 + 3x^{-2} = (y^5 + 1)^2 = y^{10} + 2y^5 + 1.$$

Thus $3x^{-2} = y^{10} + 2y^5$, so $x^2 = \frac{3}{y^{10} + 2y^5}$. Taking the positive branch gives

$$x = \sqrt{\frac{3}{y^{10} + 2y^5}}.$$

QUESTION 4

Solve $3x - 5y = 10$ and $2x + 2y = 13$.

A $(x, y) = (\frac{19}{16}, \frac{85}{16})$

B $(x, y) = (\frac{85}{16}, -\frac{19}{16})$

C $(x, y) = (\frac{85}{16}, \frac{19}{16})$

D $(x, y) = (-\frac{85}{16}, -\frac{19}{16})$

E No solutions possible.

WORKED SOLUTION

The easiest way is to double the first equation and triple the second to get:

$$6x - 10y = 20 \text{ and } 6x + 6y = 39.$$

Subtract the first from the second to give: $16y = 19$.

Therefore, $y = \frac{19}{16}$.

Substitute back into the first equation to give $x = \frac{85}{16}$.

QUESTION 5

The two inequalities $x + y \leq 3$ and $x^3 - y^2 < 3$ define a region on a plane. Which of the following points is inside the region?

A (2, 1)

B (2.5, 1)

C (1, 2)

D (3, 5)

E (1, 2.5)

F None of the above.

WORKED SOLUTION

This is fairly straightforward; the first inequality is the easier one to work with: options B, D and E violate it, so we just need to check A and C in the second inequality.

C: $1^3 - 2^2 < 3$, but A: $2^3 - 1^2 > 3$.

M2-06

Mock Test 3 - Mathematics 2

Answer **B**

QUESTION 6

How many times do $y = x + 4$ and $y = 4x^2 + 5x + 5$ intersect?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Whilst this can be done graphically, it is quicker to do algebraically. Intersections occur where the curves have the same coordinates.

Thus: $x + 4 = 4x^2 + 5x + 5$.

Simplify: $4x^2 + 4x + 1 = 0$.

Factorise: $(2x + 1)(2x + 1) = 0$.

Thus, the two graphs only intersect once at $x = -\frac{1}{2}$.

M2-07

Mock Test 3 - Mathematics 2

Answer **D**

QUESTION 7

How many times do $y = x^3$ and $y = x$ intersect?

A 0

B 1

C 2

D 3

E 4

WORKED SOLUTION

Answer: D

It's better to do this algebraically as the equations are easy to work with and you would need to sketch very accurately to get the answer. Intersections occur where the curves have the same coordinates. Thus: $x^3 = x$.

$$x^3 - x = 0$$

$$\text{Thus: } x(x^2 - 1) = 0$$

Spot the 'difference between two squares': $x(x + 1)(x - 1) = 0$

Thus there are 3 intersections: at $x = 0, 1$ and -1 .

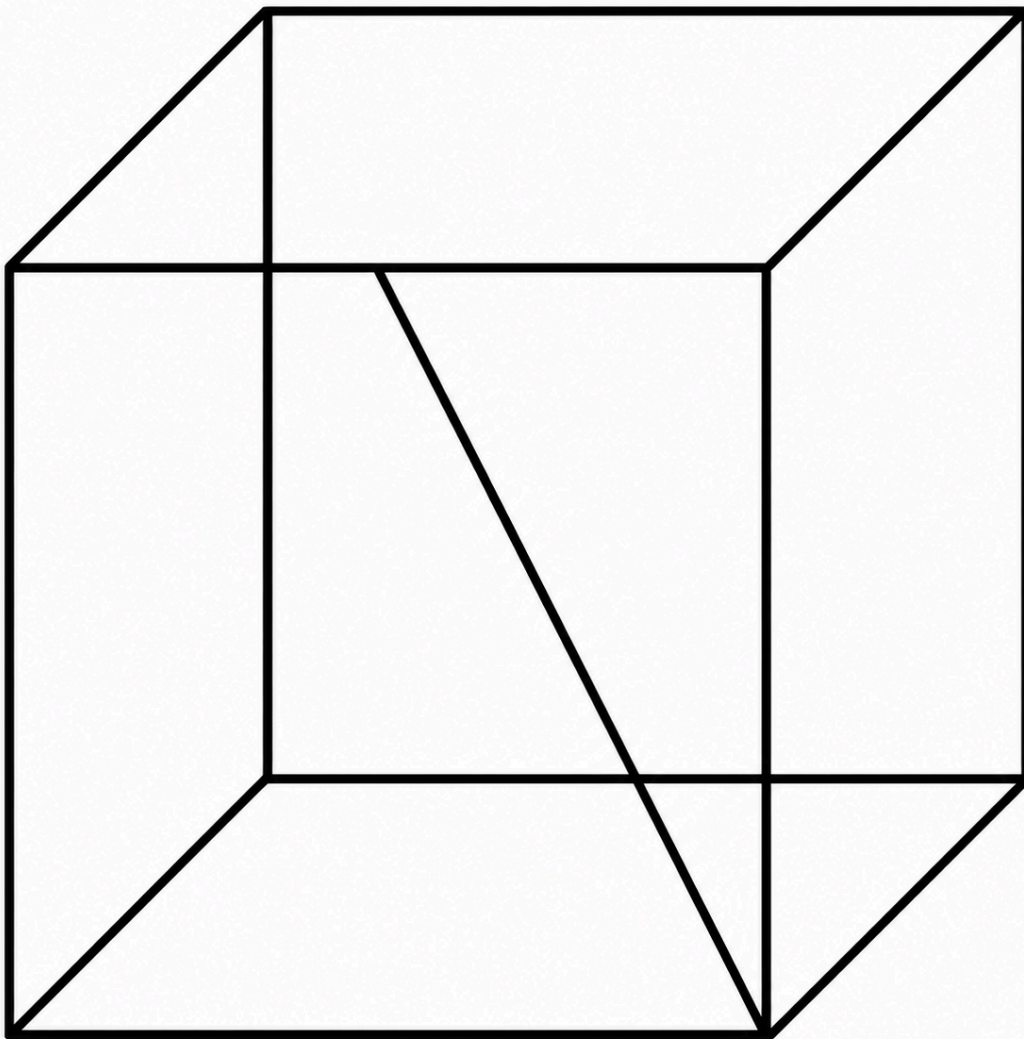
M2-08

Mock Test 3 - Mathematics 2

Answer E

QUESTION 8

A cube has unit length sides. What is the length of a line joining a vertex to the midpoint of the opposite side?



A $\sqrt{2}$

B $\sqrt{\frac{3}{2}}$

C $\sqrt{3}$

D $\sqrt{5}$

E $\frac{\sqrt{5}}{2}$

WORKED SOLUTION

Answer: E

Note that the line is the hypotenuse of a right angled triangle with one side unit length and one side of length $\frac{1}{2}$.

By Pythagoras, $(\frac{1}{2})^2 + 1^2 = x^2$.

Thus, $x^2 = \frac{1}{4} + 1 = \frac{5}{4}$.

$$x = \sqrt{\frac{5}{4}} = \frac{\sqrt{5}}{2}.$$

M2-09

Mock Test 3 - Mathematics 2

Answer D

QUESTION 9

Find the value of

$$\int_1^4 \frac{3 - 2x}{x\sqrt{x}} dx$$

A $-\frac{13}{2}$

B $-\frac{85}{16}$

C $-\frac{13}{8}$

D -1

E $-\frac{1}{4}$

F $\frac{7}{4}$

G 7

WORKED SOLUTION

Answer: D

$$\int_1^4 \frac{3-2x}{x\sqrt{x}} dx$$

(in terms of powers of x and split up the fraction)

$$= \int_1^4 \left(\frac{3}{x^{3/2}} - \frac{2}{x^{1/2}} \right) dx = \int_1^4 (3x^{-3/2} - 2x^{-1/2}) dx$$

$$= \left[\frac{3x^{-1/2}}{-1/2} - \frac{2x^{1/2}}{1/2} \right]_1^4$$

$$= [-6x^{-1/2} - 4x^{1/2}]_1^4$$

$$= \left(-6 \cdot \frac{1}{2} - 4 \cdot 2\right) - (-6 \cdot 1 - 4 \cdot 1)$$

$$= -11 - (-10)$$

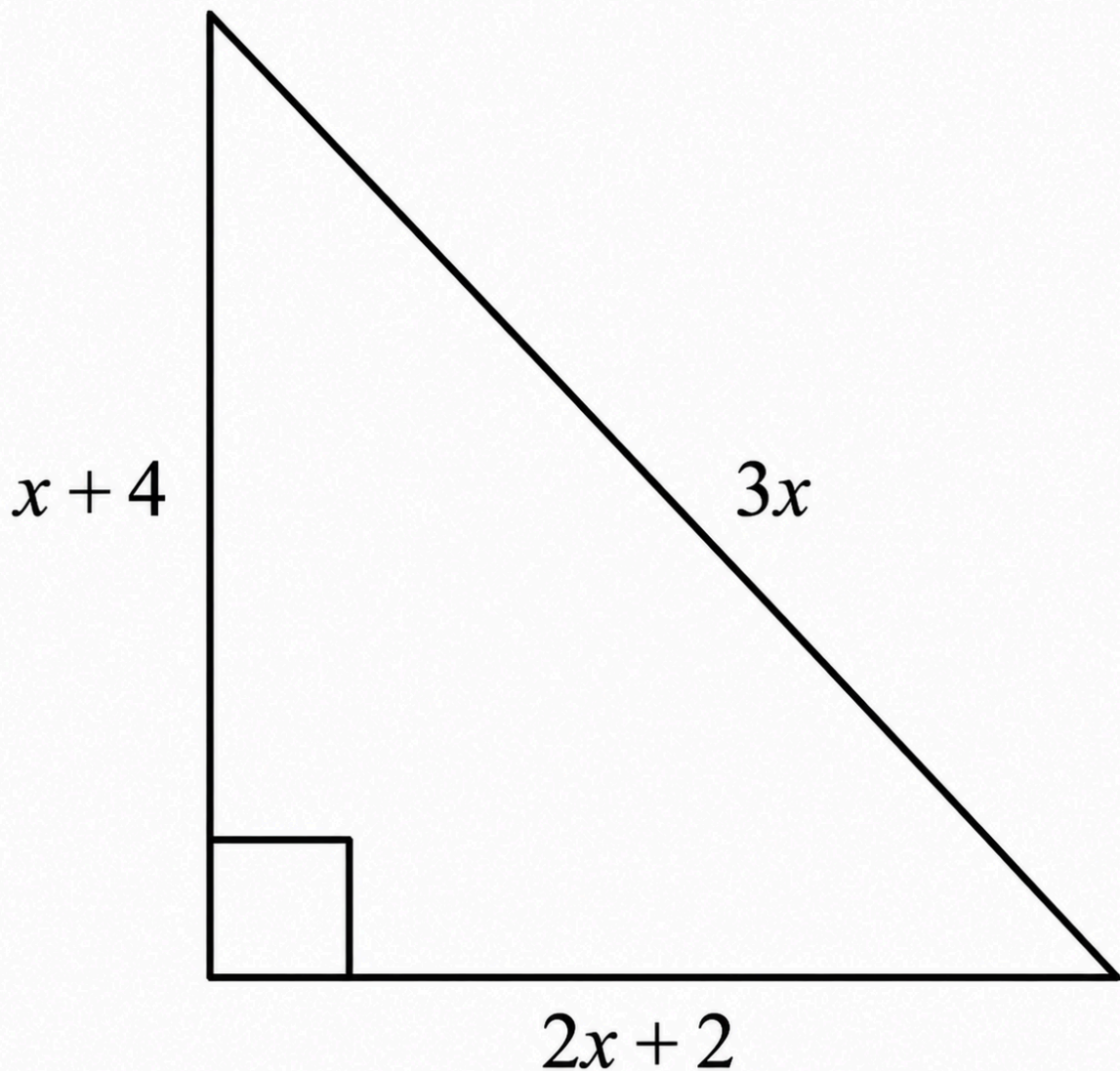
$$= -1$$

QUESTION 10

The diagram shows a right-angled triangle, with sides of length $x + 4$, $2x + 2$ and $3x$, all in cm.

[diagram not to scale]

What is the area, in cm^2 , of the triangle?



[diagram not to scale]

B 12

C 28

D 36

E 40

F 54

G 70

WORKED SOLUTION

Using Pythagoras' Theorem, $(3x)^2 = (x + 4)^2 + (2x + 2)^2$.

Thus

$$9x^2 = x^2 + 8x + 16 + 4x^2 + 8x + 4 \Rightarrow 4x^2 - 16x - 20 = 0 \Rightarrow x^2 - 4x - 5$$

Factorising: $(x - 5)(x + 1) = 0 \Rightarrow x = -1, 5$. Since $x > 0$, take $x = 5$.

$$\text{Area} = \frac{1}{2}(x + 4)(2x + 2) = \frac{1}{2}(9)(12) = 54 \text{ cm}^2.$$

M2-11

Mock Test 3 - Mathematics 2

Answer **D**

QUESTION 11

Which of the following is a simplification of $4 - \frac{x(3x + 1)}{x^2(3x^2 - 2x - 1)}$?

A
$$\frac{12x^3 - 8x^2 - 7x - 1}{x(3x - 1)(x - 1)}$$

B $\frac{4x^2 + 4x - 1}{x(x + 1)}$

C $\frac{4x^2 + 4x + 1}{x(x + 1)}$

D $\frac{4x^2 - 4x - 1}{x(x - 1)}$

E $\frac{4x^2 - 4x + 1}{x(x - 1)}$

F $\frac{12x^3 - 8x^2 - x + 1}{x(3x - 1)(x - 1)}$

WORKED SOLUTION

In the denominator, $3x^2 - 2x - 1 \equiv (3x + 1)(x - 1)$.

Considering the whole expression: $4 - \frac{x(3x + 1)}{x^2(3x + 1)(x - 1)}$.

Cancel $(3x + 1)$: $4 - \frac{1}{x(x - 1)}$.

Combine over the common denominator: $\frac{4x(x - 1) - 1}{x(x - 1)} = \frac{4x^2 - 4x - 1}{x(x - 1)}$.

M2-12

Mock Test 3 - Mathematics 2

Answer **B**

QUESTION 12

The ball for a garden game is a solid sphere of volume 192 cm^3 .

For the children's version of the game the ball is a solid sphere made of the same material, but the radius is reduced by 25%.

What is the volume, in cm^3 , of the children's ball?

A 48

B 81

C 96

D 108

E 144

WORKED SOLUTION

$$V = \frac{4}{3}\pi r^3 \Rightarrow V \propto r^3$$

$$V_{\text{child}} = \frac{4}{3}\pi r_{\text{child}}^3$$

$$= \frac{4}{3}\pi\left(\frac{3}{4}r\right)^3 = \left(\frac{3}{4}\right)^3 V$$

$$= \frac{27}{64} \times 192 = 27 \times 3 = 81 \text{ cm}^3$$

M2-13

Mock Test 3 - Mathematics 2

Answer A

QUESTION 13

Given that

$$9^{2x-1} \times \frac{1}{27^x} = 81^x$$

what is the value of x ?

A $-\frac{2}{3}$

B $-\frac{2}{5}$

C $-\frac{1}{3}$

D $-\frac{1}{4}$

E $-\frac{1}{5}$

WORKED SOLUTION

Convert all bases to 3.

$$(3^2)^{2x-1} \times \frac{1}{(3^3)^x} = (3^4)^x$$

$$3^{4x-2} \times 3^{-3x} = 3^{4x}$$

$$3^{x-2} = 3^{4x}$$

Equating exponents: $x - 2 = 4x$

$$3x = -2$$

$$x = -\frac{2}{3}$$

M2-14

Mock Test 3 - Mathematics 2

Answer **D**

QUESTION 14

PR and QS are the diagonals of a rhombus $PQRS$.

$$PR = (3x + 2) \text{ cm}$$

$$QS = (8 - 2x) \text{ cm}$$

The area of $PQRS$ is 11 cm^2 .

What is the difference, in cm, between the two possible lengths of PR ?

A $2\frac{2}{3}$

B $4\frac{1}{2}$

C $5\frac{1}{3}$

WORKED SOLUTION

Area of rhombus = product of diagonals / 2

$$\frac{(3x+2)(8-2x)}{2} = 11$$

$$12x - 3x^2 + 8 - 2x = 11$$

$$3x^2 - 10x + 3 = 0$$

$$(3x - 1)(x - 3) = 0 \Rightarrow x = \frac{1}{3}, 3$$

Possible PR lengths: $3(3) + 2 = 11$

$$3\left(\frac{1}{3}\right) + 2 = 3$$

$$11 - 3 = 8 \text{ cm difference}$$

M2-15**Mock Test 3 - Mathematics 2****Answer E****QUESTION 15**

[diagram not to scale]

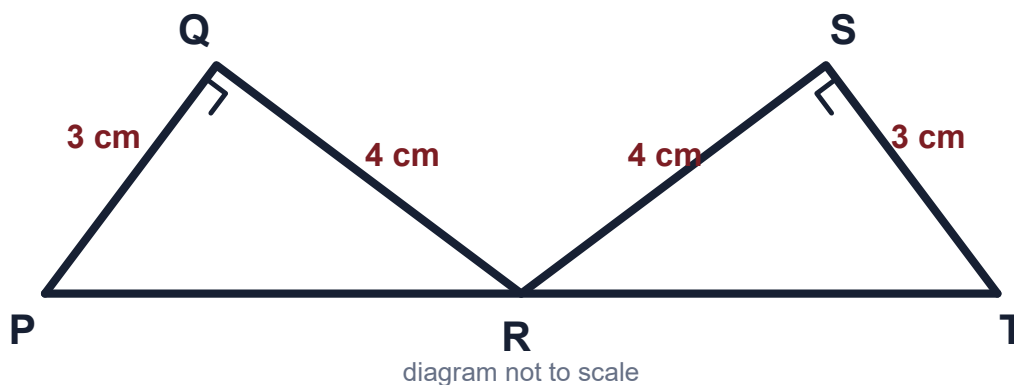
The diagram shows two congruent right-angled triangles PQR and TSR with right angles at Q and S, respectively.

$$PQ = TS = 3 \text{ cm}$$

$$QR = SR = 4 \text{ cm}$$

PRT is a straight line.

What is the length, in cm, of QS?



A 4

B $3\sqrt{2}$

C 5.2

D $4\sqrt{2}$

E 6.4

WORKED SOLUTION

Notice that the triangle PQR is similar to QUR. This can be deduced by noticing that the angles are the same. The ratio of lengths in PQR is 3:4:5 Therefore $QU = 4 \times \frac{4}{5} = 3.2\text{cm}$ $QS = 2 \times QU = 6.4\text{cm}$

M2-16

Mock Test 3 - Mathematics 2

Answer G

QUESTION 16

The total of three numbers p, q and r is 375.

The ratio p : q is 5 : 7.

The ratio $q : r$ is $4 : 11$.

What is the value of $p + r$?

A 16

B 60

C 97

D 125

E 144

F 231

G 291

H 315

WORKED SOLUTION

q can be written as $4p$.

Therefore r can be written as $11p$.

$$p + q + r = p + 4p + 11p = 16p = 256$$

Therefore, $p = 16$ and it follows that $r = 176$

$$16 + 176 = 192$$

QUESTION 17

The straight line P has equation $3y - 2x = 12$ and intercepts the y-axis at the point $(0, p)$.

The straight line Q is parallel to P, passes through the point $(6, -1)$ and intercepts the y-axis at the point $(0, q)$.

What is the value of $p - q$?

A -9

B -7

C 1

D 9

E 14

F 17

WORKED SOLUTION

To find the value of p , substitute $x = 0$ and $y = p$ into $3y - 2x = 12$

$$3p = 12$$

$$p = 4$$

The equation of P can be rearranged to $y = 23x + 4$

Therefore the gradient of P and Q is 23

Since Q passes through $(6, -1)$, the equation of Q is:

$$y - (-1) = 23(x - 6)$$

$$y = 23x - 5$$

Substitute $x = 0$, $y = q$:

$$q = -5$$

$$p - q = 4 - (-5) = 9$$

M2-18

Mock Test 3 - Mathematics 2

Answer G

QUESTION 18

The vertices of a rectangle have coordinates:

P(4, 5) Q(4, 8) R(10, 8) S(10, 5)

PQRS is transformed by a clockwise rotation of 90° about P followed by a reflection in the x-axis.

What are the coordinates of the final position of R?

A (-8, -10)

B (-7, -1)

C (-4, 1)

D (-1, 11)

E (1, -11)

F (4, -1)

G (7, 1)

H (8, 10)

WORKED SOLUTION

Relative to $P = (4, 5)$, point $R = (10, 8)$ has vector $(6, 3)$.

A clockwise 90° rotation sends $(x, y) \mapsto (y, -x)$, so $(6, 3) \mapsto (3, -6)$.

The rotated image of R is $(4, 5) + (3, -6) = (7, -1)$. Reflecting in the x-axis gives $(7, 1)$.

M2-19

Mock Test 3 - Mathematics 2

Answer C

QUESTION 19

Box A contains exactly 10 balls, of which 6 are red and 4 are blue.

Box B contains exactly 15 balls, of which 3 are red and 12 are blue.

All the balls are identical in every respect, apart from colour.

One of the two boxes is chosen at random by tossing two fair coins, as follows:

"If both coins show heads, box A is selected. Otherwise box B is selected."

One ball is then randomly taken from the selected box.

What is the probability that a red ball is taken?

A $\frac{9}{400}$

B $\frac{3}{25}$

C $\frac{3}{10}$

D $\frac{2}{5}$

E $\frac{1}{2}$

F $\frac{4}{5}$

G $\frac{323}{400}$

WORKED SOLUTION

Box A is selected only if both coins show heads, so $P(A) = \frac{1}{4}$ and $P(B) = \frac{3}{4}$.

Therefore

$$P(\text{red}) = \frac{1}{4} \cdot \frac{6}{10} + \frac{3}{4} \cdot \frac{3}{15} = \frac{3}{20} + \frac{3}{20} = \frac{3}{10}.$$

M2-20

Mock Test 3 - Mathematics 2

Answer C

QUESTION 20

Question 20:

A list of five numbers has mean x , median y and range z .

A sixth number is added to the list. This sixth number is greater than x .

Which of the following statements must be true?

- 1 The median of the six numbers cannot be one of the numbers in the list.
- 2 The mean of the six numbers is greater than x .
- 3 The range of the six numbers is greater than z .

A none of them

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Question 20: C

- Statement 1 is False as the middle two numbers could be equal.
- Statement 2 is True. Since a number larger than the mean is added, the mean will increase.
- Statement 3 is False. For the range to increase, the new number must be the new largest number; this is not necessarily stated.

M2-21

Mock Test 3 - Mathematics 2

Answer C

QUESTION 21

An arithmetic progression has first term a and common difference d .

The sum of the first 5 terms is equal to the sum of the first 8 terms.

Which one of the following expresses the relationship between a and d ?

A $a = -\frac{38}{3}d$

B $a = -7d$

C $a = -6d$

D $a = 6d$

E $a = 7d$

F $a = \frac{38}{3}d$

WORKED SOLUTION

One approach is to use the formula for the sum of the first 5 and the first 8 terms:

$$S_n = \frac{1}{2}n(2a + (n - 1)d).$$

We have

$$S_5 = \frac{1}{2} \times 5(2a + 4d) = \frac{5}{2}(2a + 4d),$$

$$S_8 = \frac{1}{2} \times 8(2a + 7d) = 4(2a + 7d).$$

These are equal, so

$$\frac{5}{2}(2a + 4d) = 4(2a + 7d).$$

Multiply both sides by 2 and expand/collect terms:

$$5(2a + 4d) = 8(2a + 7d) \Rightarrow 10a + 20d = 16a + 56d \Rightarrow -6a = 36d \Rightarrow a$$

So the answer is $a = -6d$.

M2-22

Mock Test 3 - Mathematics 2

Answer G

QUESTION 22

Consider the simultaneous equations

$$3x^2 + 2xy = 4$$

$$x + y = a$$

where a is a real constant.

Find the complete set of values of a for which the equations have two distinct real solutions for x .

A There are no values of a

B $-2 < a < 2$

C $-1 < a < 1$

D $a = 0$

E $a < -1$ or $a > 1$

F $a < -2$ or $a > 2$

G All real values of a

WORKED SOLUTION

We can solve the equations to find x by substituting for y . The second equation gives $y = a - x$, so the first equation becomes

$$3x^2 + 2x(a - x) = 4.$$

This rearranges to

$$x^2 + 2ax - 4 = 0,$$

whose discriminant is

$$(2a)^2 - 4 \cdot 1 \cdot (-4) = 4a^2 + 16.$$

This is always positive for all real a (since $4a^2 \geq 0$), so there are always two distinct real roots. Hence the correct answer is: all real values of a .

M2-23

Mock Test 3 - Mathematics 2

Answer E

QUESTION 23

Find the shortest distance between the two circles with equations:

$$(x + 2)^2 + (y - 3)^2 = 18$$

$$(x - 7)^2 + (y + 6)^2 = 2$$

A 0

B 4

C 16

D $2\sqrt{2}$

E $5\sqrt{2}$

WORKED SOLUTION

The first circle has centre $(-2, 3)$ and radius $\sqrt{18} = 3\sqrt{2}$.

The second circle has centre $(7, -6)$ and radius $\sqrt{2}$.

It would help to draw a quick sketch of the situation.

The distance between the centres is

$$\sqrt{((-2) - 7)^2 + (3 - (-6))^2} = \sqrt{81 + 81} = 9\sqrt{2}.$$

Subtracting the two radii ($3\sqrt{2}$ and $\sqrt{2}$) for the parts of the line lying within the circles gives $9\sqrt{2} - 3\sqrt{2} - \sqrt{2} = 5\sqrt{2}$. The correct answer is $5\sqrt{2}$.

This can also be proved by taking an arbitrary point on each circle, finding the distance between them and then minimising this, but that is more complicated than needed.

M2-24

Mock Test 3 - Mathematics 2

Answer D

QUESTION 24

The function f is defined by $f(x) = x^3 + ax^2 + bx + c$.

a, b, c take the values 1, 2 and 3 with no two of them being equal and not necessarily in this order.

The remainder when $f(x)$ is divided by $(x + 2)$ is R .

The remainder when $f(x)$ is divided by $(x + 3)$ is S .

What is the largest possible value of $R - S$?

A -26

B 5

C 7

D 17

E 29

WORKED SOLUTION

For this question, we make use of the remainder theorem: the remainder when $f(x)$ is divided by $(x - a)$ is $f(a)$.

In this case, we take $a = -2$ and $a = -3$, so

$$f(-2) = R$$

$$f(-3) = S$$

Substituting these into the given formula for $f(x)$ gives

$$f(-2) = -8 + 4a - 2b + c = R$$

$$f(-3) = -27 + 9a - 3b + c = S$$

We want the largest possible value of $R - S$, so we subtract these two equations to get

$$19 - 5a + b = R - S.$$

To get the largest possible value, we want a to be as small as possible and b to be as large as possible, so we take $a = 1$, $b = 3$ (and then $c = 2$), giving

$$R - S = 19 - 5 + 3 = 17.$$

Thus the correct answer is 17.

QUESTION 25

The non-zero constant k is chosen so that the coefficients of x^6 in the expansions of $(1 + kx^2)^7$ and $(k + x)^{10}$ are equal.

What is the value of k ?

A $\frac{1}{6}$

B 6

C $\frac{\sqrt{6}}{6}$

D $\sqrt{6}$

E $\frac{\sqrt{30}}{30}$

F $\sqrt{30}$

WORKED SOLUTION

Question 25: A

We can expand these two expressions using the binomial theorem, focusing on the term involving x^6 :

$$(1 + kx^2)^7 = \dots + \binom{7}{3} 1^4 (kx^2)^3 + \dots = \dots + \frac{7 \times 6 \times 5}{3 \times 2 \times 1} k^3 x^6 + \dots = \dots + 35k^3$$

and

$$(k + x)^{10} = \dots + \binom{10}{6} k^4 x^6 + \dots = \dots + \frac{10 \times 9 \times 8 \times 7}{4 \times 3 \times 2 \times 1} k^4 x^6 + \dots = \dots + 210k^4$$

We therefore need $35k^3 = 210k^4$, so $k = \frac{35}{210} = \frac{5}{30} = \frac{1}{6}$.

Thus the correct answer is $\frac{1}{6}$.

M2-26

Mock Test 3 - Mathematics 2

Answer B

QUESTION 26

Find the complete set of values of the constant c for which the cubic equation

$$2x^3 - 3x^2 - 12x + c = 0$$

has three distinct real solutions.

A $-20 < c < 7$

B $-7 < c < 20$

C $c > 7$

D $c > -7$

E $c < 20$

F $c < -20$

WORKED SOLUTION

Question 26: B

Sketch the cubic by finding its turning points and the y-intercept. For

$y = 2x^3 - 3x^2 - 12x + c$, differentiate:

$$\frac{dy}{dx} = 6x^2 - 6x - 12 = 6(x^2 - x - 2) = 6(x - 2)(x + 1).$$

Hence the stationary points are at $x = 2$ and $x = -1$ with coordinates $(-1, 7 + c)$ and $(2, -20 + c)$.

For $2x^3 - 3x^2 - 12x + c = 0$ to have three distinct real solutions, the turning points must lie on opposite sides of the x-axis. Since $-20 + c < 7 + c$, we need

$$-20 + c < 0 \text{ and } 7 + c > 0.$$

Thus $c < 20$ and $c > -7$. Combining gives $-7 < c < 20$.

M2-27

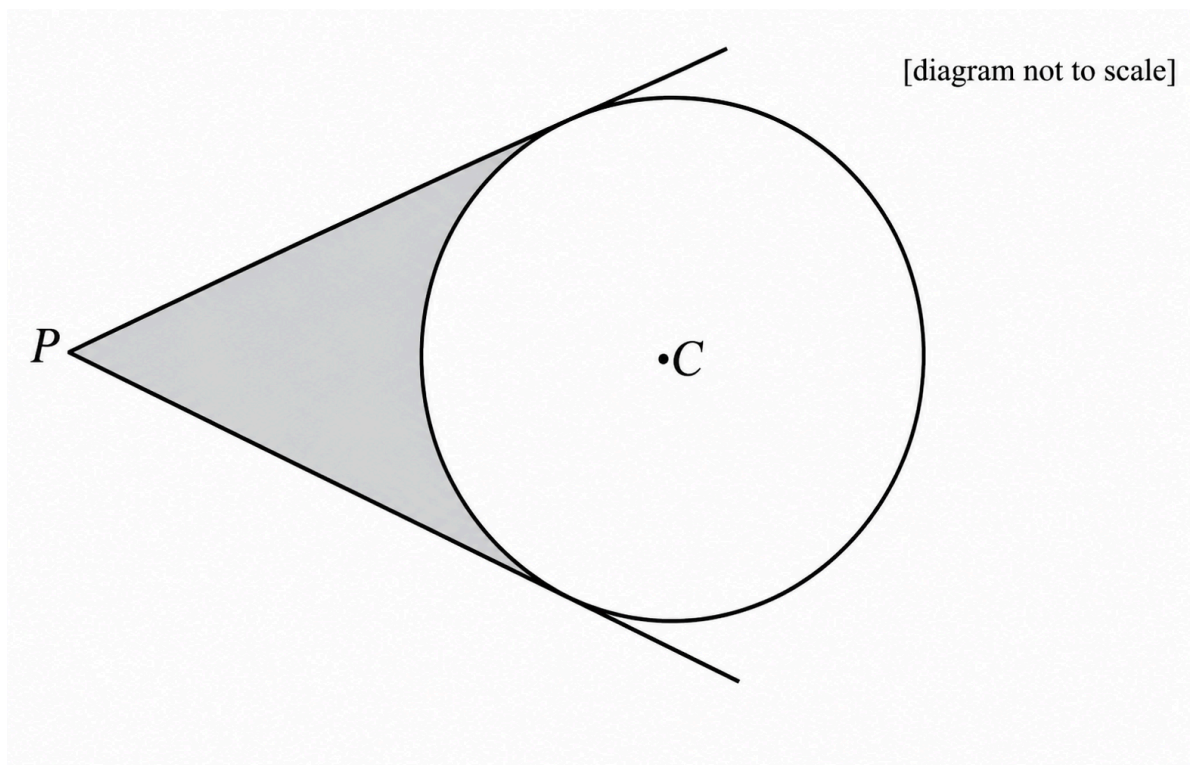
Mock Test 3 - Mathematics 2

Answer A

QUESTION 27

Tangents are drawn from a point P to a circle of radius 10 cm. The centre of the circle is C and the distance PC is 20 cm.

Which one of the following is an expression for the shaded area in square centimetres?



A $\frac{100}{3}(3\sqrt{3} - \pi)$

B $\frac{100}{3}(3\sqrt{5} - \pi)$

C $\frac{50}{3}(6\sqrt{3} - \pi)$

D $\frac{50}{3}(6\sqrt{5} - \pi)$

E $\frac{50}{3}(\sqrt{3} - 2\pi)$

WORKED SOLUTION

$$\sin \theta = 10/x = 10/20 = 1/2$$

$$\theta = 30^\circ$$

Area of the triangle:

$$2 \times \left(\frac{1}{2} \times x \cos \theta \times 10 \right)$$

$$= x\sqrt{3}/2 \times 10$$

$$= 5x\sqrt{3}$$

$$= 100\sqrt{3}$$

$$\text{Area of circular sector within triangles} = 120/360 \times \pi \times 10^2$$

$$= 100/3\pi$$

$$\text{Therefore shaded area} = 100\sqrt{3} - 100/3\pi$$

$$= \frac{100}{3}(3\sqrt{3} - \pi)$$

Chemistry

27 QUESTIONS • COMPLETE SOLUTIONS

Preserved Chemistry Standard source set. Question wording, option order, answer and solution method are unchanged.

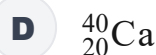
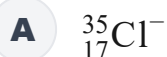
C-01

Atomic structure

Answer C

QUESTION 1

Which species has the **same number of neutrons and the same number of electrons** as ${}^{40}_{20}\text{Ca}^{2+}$?

**WORKED SOLUTION**

For ${}^{40}_{20}\text{Ca}^{2+}$, the number of neutrons is

$$40 - 20 = 20,$$

and the number of electrons is

$$20 - 2 = 18.$$

For ${}^{38}_{18}\text{Ar}$,

$$38 - 18 = 20 \text{ neutrons,} \quad 18 \text{ electrons.}$$

Both values match, so the correct answer is **C**.

C-02

Isotopes and relative atomic mass

Answer **D**

QUESTION 2

Element X occurs naturally as three isotopes of mass numbers 24, 25 and 26.

Their relative abundances are 6 : 1 : 1.

What is the relative atomic mass of X?

A 24.250

B 24.500

C 24.625

D 24.375

E 25.000

WORKED SOLUTION

Use the weighted mean:

$$A_r = \frac{6(24) + 1(25) + 1(26)}{6 + 1 + 1} = \frac{195}{8} = 24.375.$$

Therefore the correct answer is **D**.

QUESTION 3

Four atoms have the electron configurations shown.

atom	electron configuration
W	2,7
X	2,8,1
Y	2,8,7
Z	2,8,8,1

Which pair would be expected to react **most vigorously with each other** to form an ionic compound?

A W and X

B W and Z

C X and Y

D Y and Z

E W and Y

WORKED SOLUTION

W and Y are Group 17 atoms; X and Z are Group 1 atoms.

Group 1 reactivity increases down the group, so Z is more reactive than X. Group 17 reactivity decreases down the group, so W is more reactive than Y.

The most vigorous pairing is therefore the most reactive metal with the most reactive halogen: **W and Z**.

QUESTION 4

The properties of five substances are shown.

substance	melting point / °C	conducts when solid	conducts when molten
A	-20	no	no
B	660	good	good
C	805	no	good
D	1710	no	no
E	-114	no	no

Which substance is most likely to have a **giant ionic structure**?

A A

B B

C C

D D

E E

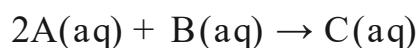
WORKED SOLUTION

A giant ionic solid normally has a high melting point. It does not conduct when solid because the ions cannot move, but it conducts when molten because the ions become mobile.

Only substance C has this combination of properties, so the answer is **C**.

QUESTION 5

The reaction



goes to completion.

Initially, $[\text{A}] = 0.80 \text{ mol dm}^{-3}$ and $[\text{B}] = 0.50 \text{ mol dm}^{-3}$, and no C is present. The volume does not change during the reaction.

What are the final concentrations of B and C?

A $[\text{B}] = 0.00, [\text{C}] = 0.50 \text{ mol dm}^{-3}$

B $[\text{B}] = 0.10, [\text{C}] = 0.20 \text{ mol dm}^{-3}$

C $[\text{B}] = 0.20, [\text{C}] = 0.40 \text{ mol dm}^{-3}$

D $[\text{B}] = 0.10, [\text{C}] = 0.80 \text{ mol dm}^{-3}$

E $[\text{B}] = 0.10, [\text{C}] = 0.40 \text{ mol dm}^{-3}$

WORKED SOLUTION

The reacting ratio is $2\text{A} : \text{B} : \text{C} = 2 : 1 : 1$.

Using all 0.80 mol dm^{-3} of A consumes

$$0.80 \div 2 = 0.40 \text{ mol dm}^{-3}$$

of B and forms the same concentration, 0.40 mol dm^{-3} , of C.

So

$$[\text{B}]_{\text{final}} = 0.50 - 0.40 = 0.10, \quad [\text{C}]_{\text{final}} = 0.40.$$

Therefore the answer is **E**.

C-06

Equilibrium

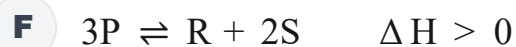
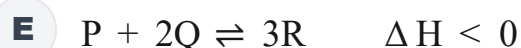
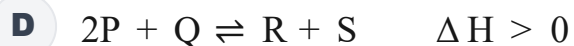
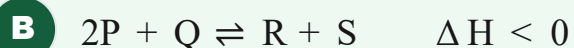
Answer **B**

QUESTION 6

For a reversible gaseous reaction:

- increasing the temperature **decreases** the equilibrium yield of R;
- increasing the pressure **increases** the equilibrium yield of R.

Which equation could represent the reaction?



WORKED SOLUTION

If increasing temperature decreases the product yield, the forward reaction must be exothermic, so $\Delta H < 0$.

If increasing pressure increases the product yield, the product side must contain fewer moles of gas.

In B there are 3 gaseous particles on the left and 2 on the right, and the forward reaction is exothermic. Therefore **B** is possible.

QUESTION 7

Chlorine reacts with cold aqueous hydroxide ions:



Consider the statements.

1. The oxidation state of chlorine in Cl_2 is 0.
2. The oxidation state of chlorine in ClO^- is +1.
3. Cl_2 acts only as an oxidising agent in this reaction.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Elemental chlorine has oxidation state 0, so statement 1 is true.

In ClO^- , if chlorine has oxidation state x ,

$$x - 2 = -1 \Rightarrow x = +1,$$

so statement 2 is true.

Some chlorine is reduced from 0 to -1 and some is oxidised from 0 to $+1$.

Chlorine therefore acts as both an oxidising and a reducing agent, so statement 3 is false.

The answer is **E: 1 and 2 only**.

C-08

Chromatography

Answer C

QUESTION 8

Four dyes are separated using the same solvent. The solvent front moves 8.0 cm.

dye	distance moved / cm
W	2.0
X	3.6
Y	5.2
Z	6.4

In a second experiment, an unknown dye moves 3.25 cm while the solvent front moves 5.0 cm.

Which dye is the unknown most likely to be?

A W

B X

C Y

D Z

E It cannot be determined.

WORKED SOLUTION

For the unknown dye,

$$R_f = \frac{3.25}{5.0} = 0.65.$$

For dye Y,

$$R_f = \frac{5.2}{8.0} = 0.65.$$

So the unknown most likely corresponds to **Y**, answer **C**.

C-09

Empirical and molecular formulae

Answer **D**

QUESTION 9

A compound contains by mass:

- 40.0% carbon,
- 6.7% hydrogen,
- 53.3% oxygen.

Its relative molecular mass is 60.

$$A_r : \quad C = 12, H = 1, O = 16$$

What is its molecular formula?

A CH₂O

B C₂H₂O₂

C C₂H₄O

D C₂H₄O₂

E C₃H₆O₃

WORKED SOLUTION

For a 100 g sample, the amounts are approximately

$$\text{C} : \frac{40.0}{12} = 3.33, \quad \text{H} : \frac{6.7}{1} = 6.7, \quad \text{O} : \frac{53.3}{16} = 3.33.$$

Dividing by the smallest gives approximately 1 : 2 : 1, so the empirical formula is CH₂O.

Its empirical formula mass is 30. Since $60/30 = 2$, the molecular formula is



Therefore the answer is **D**.

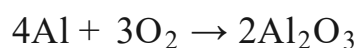
C-10

Limiting reagent and percentage yield

Answer **D**

QUESTION 10

Aluminium reacts with oxygen.



2.70 g of aluminium is mixed with 4.80 g of oxygen. The mass of aluminium oxide actually produced is 4.08 g.

$$A_r : \quad Al = 27, O = 16$$

What is the percentage yield?

A 40%

B 60%

C 75%

D 80%

E 85%

F 90%

WORKED SOLUTION

$$n(Al) = \frac{2.70}{27} = 0.100 \text{ mol}, \quad n(O_2) = \frac{4.80}{32} = 0.150 \text{ mol}.$$

Only 0.075 mol of O_2 is required for 0.100 mol of Al, so aluminium is limiting.

From the equation, 0.100 mol of Al can form 0.050 mol of Al_2O_3 .

$$M_r(Al_2O_3) = 102, \quad m_{\text{theoretical}} = 0.050(102) = 5.10 \text{ g}.$$

Thus

$$\% \text{ yield} = \frac{4.08}{5.10} \times 100 = 80\%.$$

The answer is **D**.

QUESTION 11

Solutions P and Q are both 0.10 mol dm^{-3} solutions of monoprotic acids.

- P is a strong acid.
- Q is a weak acid.

Consider the statements.

1. P has a lower pH than Q.
2. Equal volumes of P and Q require the same amount of sodium hydroxide for complete neutralisation.
3. Q has the greater electrical conductivity.

Which are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

The strong acid is more completely ionised, so it has the larger $[\text{H}^+]$ and hence the lower pH. Statement 1 is true.

Both acids are monoprotic and have the same formal concentration. Equal volumes therefore contain the same amount of acid and require the same amount of OH^- for complete neutralisation. Statement 2 is true.

The strong acid contains more mobile ions and therefore has the greater conductivity, so statement 3 is false.

The answer is **E: 1 and 2 only**.

C-12

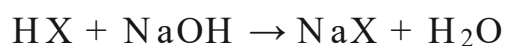
Titration

Answer **F**

QUESTION 12

A 1.56 g sample of a pure monoprotic acid HX is dissolved in water.

It requires exactly 25.0 cm^3 of 0.80 mol dm^{-3} sodium hydroxide for complete neutralisation.



What is the relative molecular mass of HX ?

A 39

B 52

C 65

D 72

E 104

F 78

WORKED SOLUTION

The amount of sodium hydroxide used is

$$n = CV = 0.80 \times 0.0250 = 0.0200 \text{ mol.}$$

The reaction ratio is 1:1, so the sample contains 0.0200 mol of HX.

Therefore

$$M_r(\text{HX}) = \frac{1.56}{0.0200} = 78.$$

The answer is **F**.

C-13

Rates of reaction

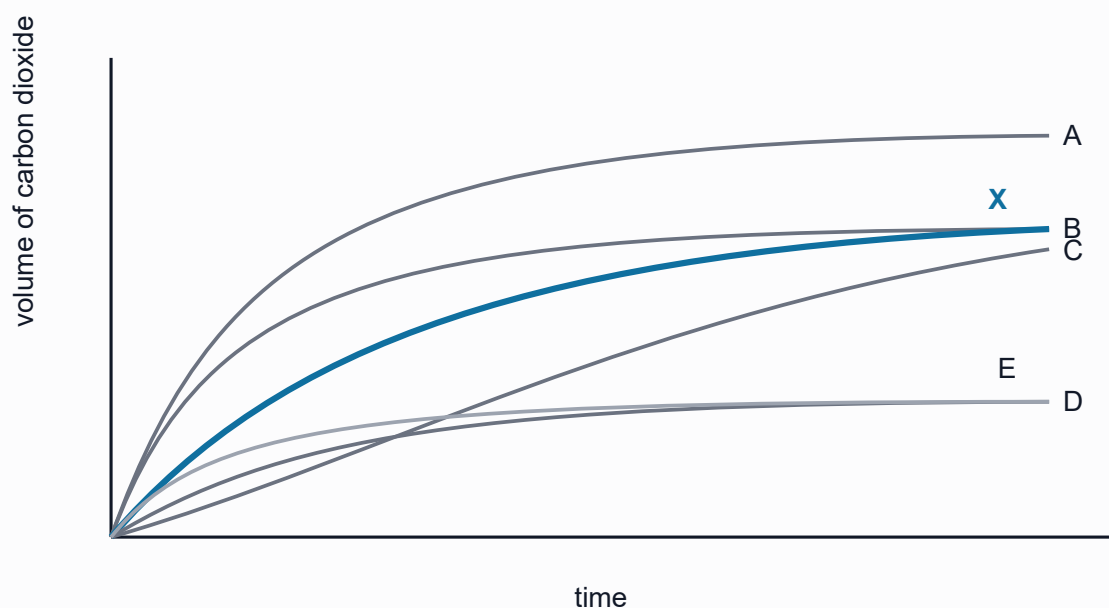
Answer **B**

QUESTION 13

Excess calcium carbonate reacts with 100 cm^3 of 1.0 mol dm^{-3} hydrochloric acid. Curve X shows the volume of carbon dioxide produced against time.

The experiment is repeated using 50 cm^3 of 2.0 mol dm^{-3} hydrochloric acid, with the same mass and particle size of excess calcium carbonate.

Which candidate curve represents the second experiment?



A curve A

B curve B

C curve C

D curve D

E curve E

WORKED SOLUTION

The amount of acid in the first experiment is

$$1.0 \times 0.100 = 0.100 \text{ mol.}$$

In the second experiment it is

$$2.0 \times 0.050 = 0.100 \text{ mol.}$$

So the same total amount of hydrochloric acid is available. Since calcium carbonate is in excess, the final amount of CO_2 is unchanged.

The second solution has twice the acid concentration, so the initial collision frequency and reaction rate are greater. The required curve must therefore reach the **same plateau more quickly**.

That is **curve B**.

C-14

Activation energy

Answer E

QUESTION 14

For a reaction,

$$\Delta H = -60 \text{ kJ mol}^{-1}.$$

The activation energy of the forward reaction is 90 kJ mol^{-1} .

What is the activation energy of the reverse reaction?

A -150 kJ mol^{-1}

B -30 kJ mol^{-1}

C 30 kJ mol^{-1}

D 90 kJ mol^{-1}

E 150 kJ mol^{-1}

F 210 kJ mol^{-1}

WORKED SOLUTION

The products are 60 kJ mol^{-1} below the reactants. The transition state is 90 kJ mol^{-1} above the reactants.

Therefore, measured from the products, the transition state is

$$90 + 60 = 150 \text{ kJ mol}^{-1}$$

higher. Hence the reverse activation energy is 150 kJ mol^{-1} , answer **E**.

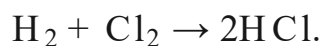
C-15

Bond energies

Answer **C**

QUESTION 15

Hydrogen reacts with chlorine:



Mean bond energies are:

bond	bond energy / kJ mol^{-1}
H-H	436
Cl-Cl	242
H-Cl	431

What is the approximate enthalpy change for the reaction?

A $- 678 \text{ kJ mol}^{-1}$

B $- 242 \text{ kJ mol}^{-1}$

C $- 184 \text{ kJ mol}^{-1}$

D $+ 184 \text{ kJ mol}^{-1}$

E $+ 678 \text{ kJ mol}^{-1}$

WORKED SOLUTION

Energy required to break the bonds in the reactants:

$$436 + 242 = 678 \text{ kJ mol}^{-1}.$$

Energy released when two H-Cl bonds form:

$$2(431) = 862 \text{ kJ mol}^{-1}.$$

Therefore

$$\Delta H = 678 - 862 = -184 \text{ kJ mol}^{-1}.$$

The answer is **C**.

QUESTION 16

Burning 0.050 mol of a fuel heats 100 g of water by 15°C.

The specific heat capacity of water is

$$4.2 \text{ J g}^{-1} \text{ }^{\circ}\text{C}^{-1}.$$

Only 70% of the energy released by the fuel is transferred to the water.

What is the enthalpy change of combustion of the fuel?

A -252 kJ mol^{-1}

B -180 kJ mol^{-1}

C -126 kJ mol^{-1}

D $+126 \text{ kJ mol}^{-1}$

E $+180 \text{ kJ mol}^{-1}$

F $+252 \text{ kJ mol}^{-1}$

WORKED SOLUTION

The energy absorbed by the water is

$$q = mc\Delta T = 100(4.2)(15) = 6300 \text{ J}.$$

This represents 70% of the energy released, so the fuel releases

$$\frac{6300}{0.70} = 9000 \text{ J} = 9.0 \text{ kJ}$$

when 0.050 mol burns.

Per mole, the magnitude is

$$\frac{9.0}{0.050} = 180 \text{ kJ mol}^{-1}.$$

Combustion is exothermic, so $\Delta H = -180 \text{ kJ mol}^{-1}$. The answer is **B**.

C-17

Electrolysis

Answer **D**

QUESTION 17

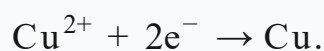
Inert electrodes are used.

Which row correctly gives the products?

- A** molten NaCl : cathode Na ; anode O_2
- B** dilute NaCl(aq) : cathode Na ; anode Cl_2
- C** $\text{CuSO}_4\text{(aq)}$: cathode H_2 ; anode Cu
- D** concentrated $\text{CuCl}_2\text{(aq)}$: cathode Cu ; anode Cl_2
- E** molten MgCl_2 : cathode H_2 ; anode Cl_2

WORKED SOLUTION

In concentrated aqueous copper(II) chloride with inert electrodes, Cu^{2+} is reduced at the cathode:



At the anode, chloride ions are oxidised:



So the correct row is **D**.

QUESTION 18

A 0.250 dm^3 solution initially contains 0.100 mol of Cu^{2+} .

Copper is electroplated from the solution until 3.20 g of copper has been deposited.

The volume is then restored to 0.250 dm^3 by adding water.

$$A_r(\text{Cu}) = 64$$

What is the final concentration of Cu^{2+} ?

A 0.10 mol dm^{-3}

B 0.15 mol dm^{-3}

C 0.20 mol dm^{-3}

D 0.25 mol dm^{-3}

E 0.40 mol dm^{-3}

WORKED SOLUTION

The amount of copper deposited is

$$n(\text{Cu}) = \frac{3.20}{64} = 0.0500 \text{ mol.}$$

Each mole of deposited copper removes one mole of Cu^{2+} , so the amount remaining is

$$0.100 - 0.0500 = 0.0500 \text{ mol.}$$

After the volume is restored to 0.250 dm^3 ,

$$[\text{Cu}^{2+}] = \frac{0.0500}{0.250} = 0.20 \text{ mol dm}^{-3}.$$

The answer is **C**.

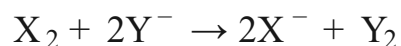
C-19

Halogen displacement

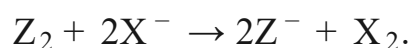
Answer **A**

QUESTION 19

The following reactions occur:



and



What is the order of reactivity of the halogens, from most to least reactive?

A $\text{Z} > \text{X} > \text{Y}$

B $\text{X} > \text{Z} > \text{Y}$

C $\text{X} > \text{Y} > \text{Z}$

D $\text{Y} > \text{X} > \text{Z}$

E $\text{Z} > \text{Y} > \text{X}$

WORKED SOLUTION

Because X_2 displaces Y^- , X is more reactive than Y:



Because Z_2 displaces X^- , Z is more reactive than X:



Therefore $\boxed{\text{Z} > \text{X} > \text{Y}}$, answer **A**.

QUESTION 20

Four metals M, Q, R, T give these observations.

1. Q displaces R from aqueous RSO_4 .
2. T also displaces R from aqueous RSO_4 .
3. T does not displace Q from aqueous QSO_4 .
4. R reacts with dilute hydrochloric acid.
5. M does not react with dilute hydrochloric acid.

Which is the correct order from most reactive to least reactive?

A $Q > R > T > M$

B $T > Q > R > M$

C $Q > T > R > M$

D $Q > T > M > R$

E $R > Q > T > M$

F $T > R > Q > M$

G $M > R > T > Q$

H $Q > M > T > R$

WORKED SOLUTION

Q displaces R, so $Q > R$. T displaces R, so $T > R$. T does not displace Q, so $Q > T$.

Thus $Q > T > R$.

R reacts with hydrochloric acid, so R is above hydrogen in the reactivity series. M does not react with hydrochloric acid, so M is below hydrogen and therefore below R.

The full order is $Q > T > R > M$, answer **C**.

C-21

Fractional distillation

Answer **B**

QUESTION 21

Hexane has a boiling point of 69°C . Octane has a boiling point of 126°C .

An approximately equal mixture of the liquids is separated using a fractionating column.

When the **first drops of distillate** are collected, which observation is most likely?

- A** flask below 69°C ; top near 69°C
- B** flask between 69°C and 126°C ; top near 69°C
- C** flask near 126°C ; top near 69°C
- D** flask between 69°C and 126°C ; top near 126°C
- E** flask near 126°C ; top near 126°C

WORKED SOLUTION

The liquid mixture in the flask boils over a temperature range lying between the boiling points of the two pure components.

Hexane is more volatile, so the vapour reaching the top of an efficient fractionating column is strongly enriched in hexane. When the first distillate is collected, the temperature near the top should therefore be close to 69°C .

Thus **B** is the most likely observation.

QUESTION 22

25 cm³ of a gaseous alkene is completely burned in excess oxygen.

100 cm³ of carbon dioxide is produced.

All gas volumes are measured at the same temperature and pressure.

Which is the molecular formula of the alkene?

A C₂H₄

B C₃H₆

C C₄H₈

D C₄H₁₀

E C₅H₁₀

WORKED SOLUTION

At the same temperature and pressure, gas volume ratios equal mole ratios.

The carbon dioxide volume is four times the alkene volume:

$$\frac{100}{25} = 4.$$

One molecule of C_nH_{2n} produces n molecules of CO₂, so n = 4.

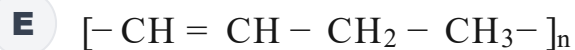
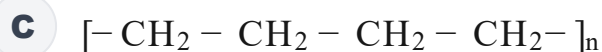
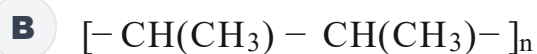
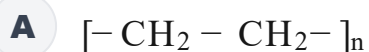
The alkene is C₄H₈, answer **C**.

QUESTION 23

But-1-ene has condensed structural formula



Which repeating unit represents poly(but-1-ene)?

**WORKED SOLUTION**

In addition polymerisation, the carbon–carbon double bond opens to form the polymer backbone. The ethyl group attached to the second alkene carbon remains as a side group.

The repeating unit is therefore



The answer is **D**.

QUESTION 24

A carboxylic acid X reacts with ethanol to form an ester Y and water.

The relative molecular mass of Y is 102.

$$M_r(\text{ethanol}) = 46, \quad M_r(\text{H}_2\text{O}) = 18.$$

What is the relative molecular mass of X?

A 46

B 60

C 74

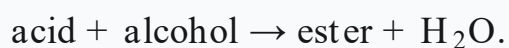
D 88

E 102

F 120

WORKED SOLUTION

Esterification is



Mass is conserved, so

$$M_r(\text{X}) + 46 = 102 + 18.$$

Hence

$$M_r(\text{X}) = 120 - 46 = 74.$$

The answer is **C**.

QUESTION 25

An aqueous solution of a salt gives:

- a **green flame**;
- a **blue precipitate** when aqueous sodium hydroxide is added;
- a **white precipitate** when acidified aqueous silver nitrate is added.

Which salt is present?

A CuCl_2

B CuSO_4

C FeCl_2

D NaCl

E CaCl_2

F $\text{Cu}(\text{NO}_3)_2$

WORKED SOLUTION

The green/blue-green flame and blue precipitate with sodium hydroxide indicate Cu^{2+} .

The white precipitate with acidified silver nitrate indicates chloride ions, because AgCl is white.

The salt is therefore $\boxed{\text{CuCl}_2}$, answer **A**.

QUESTION 26

Consider the statements.

1. Carbon monoxide can be formed by incomplete combustion of carbon-containing fuels.
2. Sulfur dioxide can be produced when sulfur-containing impurities in fuels are burned and can contribute to acid rain.
3. Nitrogen oxides are produced mainly because carbon burns with insufficient oxygen.

Which statements are correct?

A none

B 1 only

C 2 only

D 3 only

E 1 and 2 only

F 1 and 3 only

G 2 and 3 only

H 1, 2 and 3

WORKED SOLUTION

Statement 1 is correct: incomplete combustion can produce carbon monoxide.

Statement 2 is correct: sulfur-containing impurities can form sulfur dioxide, which contributes to acid rain.

Statement 3 is false. Nitrogen oxides are formed when nitrogen and oxygen react at high temperatures, for example in combustion engines; they are not mainly a consequence of carbon burning with insufficient oxygen.

Therefore the answer is **E: 1 and 2 only**.

C-27

Quantitative chemistry

Answer C

QUESTION 27

A 6.00 g mixture contains only magnesium and magnesium oxide.

The mixture reacts completely with excess dilute hydrochloric acid.

Exactly 2.40 dm^3 of hydrogen gas is produced at room temperature and pressure.

At rtp,

$$1 \text{ mol gas} = 24.0 \text{ dm}^3.$$

$$A_r(\text{Mg}) = 24$$

What percentage by mass of the original mixture was magnesium?

A 20%

B 30%

C 40%

D 50%

E 60%

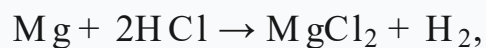
WORKED SOLUTION

Only magnesium metal produces hydrogen with dilute hydrochloric acid.
Magnesium oxide neutralises the acid but does not form hydrogen.

The amount of hydrogen formed is

$$n(\text{H}_2) = \frac{2.40}{24.0} = 0.100 \text{ mol.}$$

From



the amount of magnesium is also 0.100 mol. Its mass is

$$0.100(24) = 2.40 \text{ g.}$$

Therefore

$$\% \text{Mg} = \frac{2.40}{6.00} \times 100 = 40\%.$$

The answer is **C**.

Compilation record: Level 1 uses Engineering Mock Test 3 for Mathematics/Physics and the corresponding Standard Biology/Chemistry source sets. This solution book is ready to support the later construction of the matching full test.